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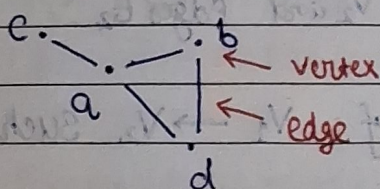
Graph Theory

Simple graphs

G consists of two sets: ① Non empty V : Set of vertices
② E : Set of two element subsets of V
called edges

example: $V = \{a, b, c, d\}$

$E = \{\{a, b\}, \{a, c\}, \{a, d\}, \{b, d\}\}$



Defⁿs:

- ① Adjacent vertices: if they are joined by an edge.
- ② An edge is said to be incident on the vertices it joins
- ③ No. of ~~vertices~~ edges incident on a vertex is called its degree.

Properties of Simple graphs:

- ① No self loop
- ② At most one edge b/w two vertices.
- ③ At least one vertex, but there may not be any edge.

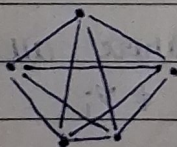
Handshaking Lemma

The sum of the degree of vertices in a graph equals twice the no. of edges.

Some common graphs:

- ① Complete graph on n vertices K_n
- ② Empty graph

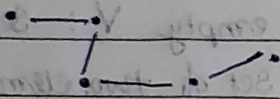
K_5 :



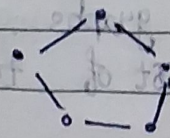
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③ Line graph L_4



④ Simple Cycle C_5



Graph Isomorphism:

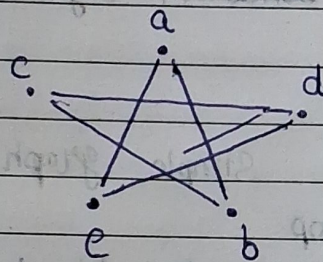
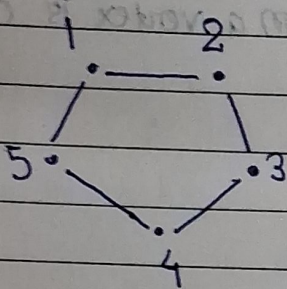
G_1 : graph with vertices V_1 and edges E_1

G_2 : graph with vertices V_2 and edges E_2

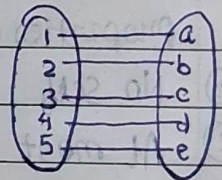
$G_1 \cong G_2$ iff $\exists$ a bijection $f: V_1 \rightarrow V_2$ such that
 $u, v \in V_1$:

$$u-v \in E_1 \Leftrightarrow f(u)-f(v) \in E_2$$

example:



f :



problems: Pg - 174 to 178

Path in a graph G :

Sequence of vertices $v_0, v_1, v_2, \dots, v_k$ with $k \geq 0$

such that $v_i - v_{i+1}$ is an edge for $0 \leq i < k$

• path is said to start at v_0 and end at v_k

and length of path is defined to be k .

Simple path: a path where all v_i 's are different
i.e. $i \neq j \Rightarrow v_i \neq v_j$

Note: v_0 : path with length 0

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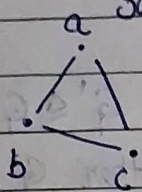
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Shift terminology:

path $\equiv$ walk

Simple path $\equiv$ path

Cycle: described by a path that begins and ends at the same vertex



abca

bcab

cabc

} more than one path can describe cycle

Simple Cycle: one that doesn't cross or backtrack on itself.

length of a simple cycle = no. of distinct vertices in it
 except

Single vertex: cycle of length 0

cycle of length 1 or 2 not possible

Defⁿ: (G', E', V') is said to be subgraph of (G, E, V) if $E' \subseteq E$ and $V' \subseteq V$ and G' satisfies graph properties

Defⁿ: for $n \geq 3$ C_n be graph with $V = [n]$ and $E \in \{1-2, 2-3, \dots, (n-1)-n, n-1\}$

Simple Cycle of length n : A graph $\cong C_n$ for some $n \geq 3$
 a Simple cycle in G is a subgraph of $G \cong C_n$

Conn Connected Components:

Two vertices u and v are connected in G if $\exists$ a path that begins at u and ends at v

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Connected graph: when every pair of vertices are connected.

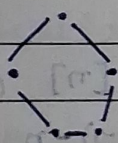
A connected component in a graph G is a maximal connected subgraph of G

$G' \trianglelefteq G$ i.e. $\nexists G''$ such that $G' \trianglelefteq G'' \trianglelefteq G$
then either $G' = G''$
or $G'' = G$

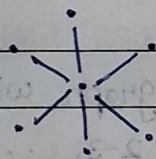
K edge connected graph:
Two vertices in a graph G are K edge connected if they remain connected in every subgraph obtained by deleting $K-1$ edges.

A graph with at least two vertices is K -edge connected if every two of its vertices are K -edge connected.

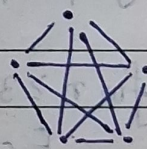
Ex: 



$K=2$



$K=1$



$K=4$

$K_n = n-1$

$(n-1)$ is degree of each vertex

K -connected $\Rightarrow i$ -connected where $i \leq K$

Theorem:

if two vertices are K edge connected
 $\Leftrightarrow K$ edge-disjoint paths connecting them

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Lemma: if vertex u is connected to vertex v in a graph, then there is a simple path from u to v .

proof: since u and v are connected

if exist a path from u to v

if path length = 0 then $u = v$ which indeed a simple path

Suppose path length ≥ 1

all possible path lengths $\subseteq \mathbb{N} \setminus \{0\}$

$\therefore$ by well ordering principle

if a min path from u to v of minimum length,

let it be $v_0 v_1 \dots v_k$

$v_0 = u \quad v_k = v \quad k \geq 2$

Claim: this a simple path

proof by contradiction: Assuming it's not a simple path

i.e $\exists v_i = v_j \quad 0 \leq i < j \leq k$

$v_i = v_j$

Then deleting $v_i, v_{i+1}, \dots, v_j$

gives a strictly shorter path: $v_0 v_1 \dots v_i v_{i+1} \dots v_k$ from u to v

$(\Rightarrow \Leftarrow) \therefore v_0 v_1 \dots v_i v_{i+1} \dots v_k$

was ~~sim~~ shortest path

Since assumption resulted in

contradiction $\therefore v_0 v_1 \dots v_k$

indeed a simple path

~~Corollary~~

Corollary: for any path of length k in a graph, there is a simple path of length at most k with the same endpoints.

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Theorem:

Every graph with v vertices and e edges has at least $v - e$ connected components.

Proof: By induction on the no. of edges e .

$P(e)$: $\forall v$ every graph with $|V| = v$ $|E| = e$ has at least $v - e$ connected components.

Basis: $P(0)$: no edges

All vertices are components $= v$

$$\therefore v - e = v$$

$v \geq v \therefore$ true for $e = 0$.

Inductive step: Assume $P(e)$ is true.

consider G with $(e+1)$ edges and v vertices.

Suppose we removed an arbitrary edge $a-b$ and call the graph G'

we know # connected components in $G' \geq (v - e)$

now add $a-b$ back into G'

Case I: if $a-b \in$ different components of G'

i.e. two components merged into 1 in G'

$$\therefore \# \text{ CC in } G \geq \# \text{ CC in } G' - 1 \geq v - e - 1$$

$$= v - (e+1)$$

Case II: $\# \text{ CC in } G = \# \text{ CC in } G' \geq v - e > v - (e+1)$

$P(e+1)$ holds, in either case.

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Corollary: Every connected graph G with V vertices has at least $V-1$ edges

$$\# \text{ cc. in } G \geq V - e$$

$$e \geq V - \# \text{ cc. } G = V - 1$$

$$\therefore e \geq V - 1$$

Problems: Pg 183 to Pg 187

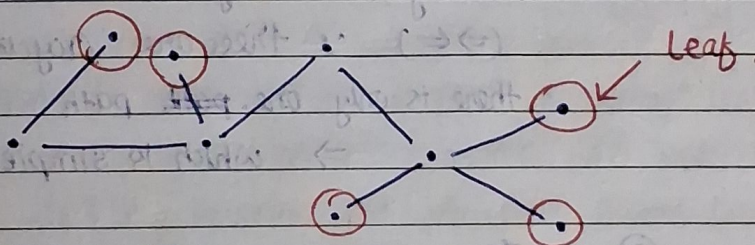
Trees

A graph without any simple cycle is called acyclic

A connected acyclic graph is called tree

Note: trees can be rooted or non-rooted

Example:



leaf: vertex of degree at most one.

Note: ~~for~~ there is only one case where a vertex can have degree = 0 is graph with single vertex

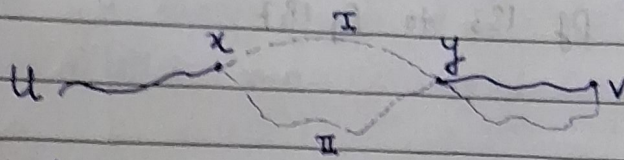
Theorem: (properties of Tree)

- ① Any connected subgraph is a tree.
- ② There is a unique simple path b/w any two pair of vertices.
- ③ Adding an edge b/w two vertices create a cycle.
- ④ Removing any edge disconnects the graph.
- ⑤ if it has at least two vertices, then it has at least two leaves
- ⑥ The no. of vertices is one larger than the number of edges.

Proof: ① A simple cycle in subgraph is simple cycle in graph. If subgraph is also connected then it's defⁿ of tree.

② $\because$ graph is connected $\therefore$ there is at least one path from u to v .

Suppose there are two different paths from u to v .



Let x be the first vertex where the both paths diverge.

and let y be the next vertex they share.

then the path I & II from x to y have no common edges $\Rightarrow$ they combined form a cycle.

$(\Rightarrow \Leftarrow) \because$ there are acyclic.

$\therefore$ there is only one path.

$\Rightarrow$ which is simple path.

③ $\because$ If a unique path from u to v adding an edge $u-v$ results in simple cycle.

④ Suppose we removed $u-v$.

$\because$ this edge was the unique path from u to v .

$\therefore u$ & v are not connected now.

$\therefore$ the graph becomes disconnected.

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(5). Consider the longest simple path $v_1, \dots, v_m$

(i) there can't be any edge $v_1 - v_i$, $2 \leq i \leq m$

otherwise it will form a cycle.

(ii) there can't be any edge $u - v_1$, otherwise it will

increase the path length but we already have longest one

$\therefore v_1 - v_2$ is only edge with v_1

$\therefore v_1$ is a leaf

Similarly v_m is a leaf

} so we found two leaf leaves
more are possible.

(6) Induction of no. of vertices.

$P(v)$: a tree ^{with} v vertices has $v-1$ edges

$P(1)$: only one vertex $\Rightarrow$ no edge

indeed true.

Suppose $P(v)$ is true. Claim: $P(v+1)$ is true

we know for $v \geq 2$ $\exists$ at least two leaves ^{let T be a tree with $v+1$ vertices}

now consider removing the edge connecting ^{one of the} leaf to tree

it results in two connected components T' and $\{v\}$ the leaf vertex
($\because$ leaf only have one edge $\therefore$ second component contains only it)

T' is a tree ($\because$ rest of the vertices are connected and acyclic)

T' has v vertices

$\therefore$ it must have $v-1$ edges

not adding the deleted edge again make T' a T with v edges

$T(v+1)$ is true

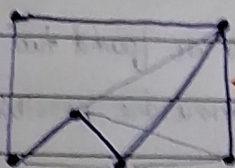
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Spanning Tree for a Graph G

Consider a subgraph G' of G

if G' is a tree then it is called a spanning tree if it contains all the vertices of G .



← Spanning tree

Theorem: Every connected graph consists a spanning tree.
proof:

Let T be a connected subgraph of G with same vertices as G with smallest no. of edges possible.

Claim: T is acyclic

Suppose T is cyclic with cycle $v_0 - v_1, v_1 - v_2, \dots, v_n - v_0$

Suppose we remove $v_n - v_0$

Suppose there are two vertices x & y

if they are joined by path containing $v_n - v_0$ they are still joined

if not they are joined by path containing remainder cycle.

$\therefore$ deleting this edge doesn't result in disconnecting graph but T has minimum edges.

$(\Rightarrow \Leftarrow)$

$\therefore$ there can't be any cycle present.

Hence T is the spanning tree.

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problem: Pg - 190 to 192.

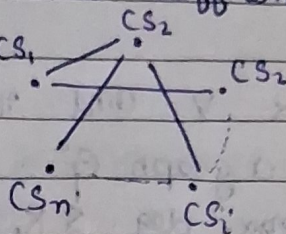
Coloring Graphs

-coloring the nodes of graph.

Valid Coloring:

given a Graph G assign a colour to each node such that ~~adjacent~~ adjacent nodes have different colors.

example: scheduling:



Defⁿ:

A Graph is k -colourable if it has a colouring that uses at most k colours.

K_n : n colourable but not $(n-1)$ colourable

C_n : if n is even $\rightarrow$ 2 colourable

n is odd $\rightarrow$ 3 colourable

Chromatic Number:

The minimum ^{value} ~~no~~ of k for which a graph G has a valid coloring is called its chromatic number $\chi(G)$

Theorem: A graph with max degree at most k is $(k+1)$ colourable.

proof:

Induction on number of vertices n .

$P(n)$ be ~~prop~~ proper

$P(n)$: n vertex graph with max^m degree at most k is $(k+1)$ colorable.

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Base case ($n=1$)

max-degree = 0 and it is 1-colourable.

$P(1)$ is true.

Induction: Suppose $P(n)$ is true.

Let G be an $(n+1)$ vertex graph with max degree at most K .

Take a vertex v and remove all edges incident on v then we get a graph G' .

G' has max-deg $\leq K$.

By IH G' is $(K+1)$ colourable.

now reattach v to G' to make it G

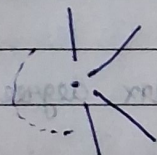
v has at most K neighbours and we have $K+1$ colours.

$\therefore v$ can be assigned ~~$K+1$~~ different colour (different from all adjacent) but already in Graph

$\therefore G$ is $(K+1)$ Colourable.

$$\chi(K_n) = n$$

Star graph :



max degree = $n-1$

$$\chi(G) = 2$$

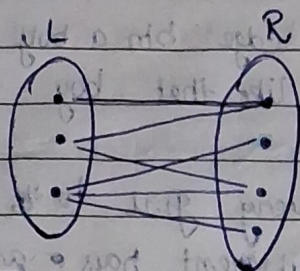
problem - pg-195 to 198

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Bipartite Graphs

The vertices can be partitioned into two sets L and R , such that every edge has one end point in L and the other in R .



fact: A graph is bipartite iff it is 2 colourable.

Theorem: A graph is bipartite iff it has no odd-length cycle.

proof: ($\Rightarrow$) Suppose G is bipartite and $v_1, \dots, v_n, v_1$ is a cycle in G .

Suppose $v_1 \in L$, Then $v_2 \in R, v_3 \in L, \dots$

$v_n \in R$ if n is even

$v_n \in L$ if n is odd

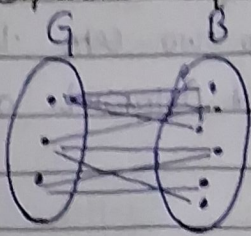
but v_1, v_n is an edge $\therefore v_n \notin L \therefore n$ can't be odd

$\therefore$ only even length cycles possible

($\Leftarrow$)

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Bipartite Matchings



$|B| \leq |G|$

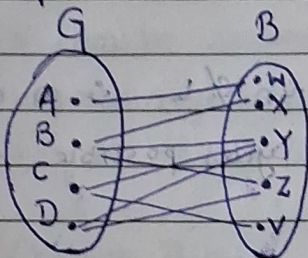
each girl like some boys and don't like others.

there is an edge b/w a boy and girl iff girl like that boy.

matching: A way of assigning every girl to a boy so that different girls are assigned different boys and a girl is assigned to a boy she likes.

Defⁿ: The set of boys liked by a set of girls consists of all the boys liked by at least one of those girls.

Example:



Set of boys
 $\{X, Y, Z, V\}$ liked by $\{B, C\}$
Set of girls

Matching condition: Every subset of girls likes at least as ~~large~~ large as set of boys.

Theorem: A matching for a set of girls G with a set of boys B can be found iff matching conditions holds

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proof: ($\Rightarrow$) Suppose a matching exists

Suppose a subset $X \subseteq G$ and let $Y \subseteq B$ be the set of boys liked by X

if $|X| > |Y|$ then some girls in X can't be paired with a boy she like

$$\therefore |X| \leq |Y|$$

($\Leftarrow$) considering matching condition to be true
strong induction on $|G|$

Base Case: ($|G| = 1$)

the matching condition $\Rightarrow$ lone girl likes at least one boy so a matching exist.

Inductive step: $|G| \geq 2$ for the bipartite graph say H_0

$\therefore |G| > 1$ we take one of the girl say g_1 and a boy liked by her say b_1 pair them up and remove them from graph to get a new bipartite graph H_1 .

Case-I: if Hall's condition holds for H_1

then by IH $\exists$ a matching for H_1

this matching $U = \{(g_1, b_1)\}$

$U \cup$ is the matching for G

Hence IH holds true

Case-II: if Hall's condition doesn't hold for H_1

$\therefore \exists X_1 \subseteq G - \{g_1\}$ and $Y_1 \subseteq B - \{b_1\}$ liked by X_1 in H_1

$$\text{and } |X_1| > |Y_1|$$

now $X \subseteq G$ so in H_0 matching condition for X holds.

let Z be set of boys liked by X in H_0

$$Z \subseteq B$$

So matching condition $\Rightarrow |X| \leq |Z|$

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So $|X| \leq |Z|$ and just by removal of b_1 , $|X| > |Y|$

$\therefore |X| = |Z|$ earlier, and $b_1 \in Z$.

now let's make another bipartite subgraph of H_0

Say H_2

by removing X from G and all boys liked by X i.e. Z from B .

H_2 is bipartite from $G - X \rightarrow B - Z$

now let's consider $X' \subseteq G - X$

and $Z' \subseteq B - Z$

Such Z' is set liked by X' in H_2

clearly X & X' are disjoint $\therefore |X \cup X'| = |X| + |X'| \dots (2)$

now in H_0 $|X \cup X'| \leq |H_0|$

matching condition ~~not~~ holds for them

~~that~~ $|X \cup X'| \leq$ set of all boys liked by $X \cup X'$ in G

$\Rightarrow |X| + |X'| \leq |Z| + |Z'|$

$\Rightarrow |X'| \leq |Z'|$

i.e., matching condition holds for X' in H_2 .

So, in H_0 let's take X matching conditions

holds for X $\therefore |X| \leq |H_0|$ by IH

If a matching for X say M_1

match all girls in X and remove them

from H_0 also all boys liked by

X and Z from B to get H_2

now we just proved $|X'| \leq |Z'|$ matching condition

~~holds~~ holds. If a matching for $G - X$ say M_2

now $M_1 \cup M_2$ is the matching for entire H_0 .

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Formal Definitions

- ① A matching in a graph G is a set of edges such that no pair of edges share a vertex.
- ② A matching M is said to cover a set L of vertices if each vertex in L is an endpoint of an edge in M .
- ③ In any graph the set $N(S)$ is the set of neighbours of the vertices in S is the set of all vertices adjacent to some vertex in S .
$$N(S) = \{ v \mid u-v \text{ is an edge for some } u \in S \}$$
- ④ A set S of vertices is called a bottle neck if $|S| > |N(S)|$.

Hall's Theorem: Let G be a bipartite graph with vertex partitions L and R . Then there is a matching in G that covers L iff no subset of L is a bottle neck.

Defⁿ: A bipartite graph is called degree constraint if $\deg(v) \geq \deg(w) \forall w \in L \text{ and } v \in R$.
 L, R are vertex partitions.

Lemma: Every degree constraint bipartite graph satisfies the matching condition.

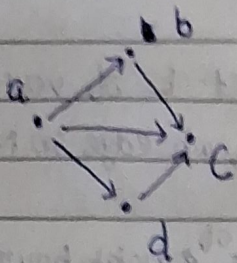
Problems: Pg 203 to 206.

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Directed Graphs

$G = (V, E)$ where $E \subseteq V \times V$

Example:



$V = \{a, b, c, d\}$

$E = \{(a, b), (b, c), (a, c), (a, d), (d, c)\}$

Source/ tail target/ head
 $u \xrightarrow{e} v$

directed edge:

$e = \langle u \rightarrow v \rangle$

indegree $(v) = |\{e \in E(G) \mid e = (u, v) \text{ for some } u\}|$

outdegree $(v) = |\{e \in E(G) \mid e = (v, w) \text{ for some } w\}|$

Lemma: $\sum_{u \in V} \text{indeg}(u) = \sum_{u \in V} \text{outdeg}(u)$

path: A nonempty sequence of vertices $v_0, v_1, \dots, v_n$

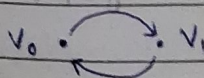
such that $(v_i, v_{i+1}) \in E$ for $0 \leq i \leq n-1$

- Start at v_0 and end at v_n

- length of the path is n

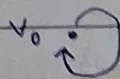
Simple path: All v_i 's are distinct.

Closed path: begins and ends at the same vertex cycle. They are always of even length.



v_0, v_1, v_0
 2 length

length one cycles are possible



Self loop

$E \subseteq V \times V$

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Theorem: The shortest path from any vertex to another vertex is a simple path (no repetition of vertices)

Defⁿ: The distance $\text{dist}(u, v)$ in a digraph is the length of shortest path from u to v
if there is no path from u to v $\text{dist}(u, v) = \infty$

Lemma: $\text{dist}(u, v) \leq \text{dist}(u, w) + \text{dist}(w, v)$
for $u, v, w \in E$

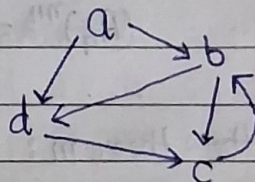
Adjacency Matrix

If a graph G has n vertices $v_0, v_1, \dots, v_{n-1}$. Then G can be represented as $n \times n$ matrix of zeroes and ones called adjacency matrix A_G .

$$(A_G)_{ij} = \begin{cases} 1 & \text{if } \langle v_i \rightarrow v_j \rangle \in E(G) \\ 0 & \text{otherwise} \end{cases}$$

example:

	a	b	c	d
$A_H = a$	0	1	0	1
b	0	0	1	1
c	0	1	0	0
d	0	0	1	0



Application:

The matrix $(A_G)^k$ provides the count of length k paths b/w any two vertices. $k \geq 0$

$(A_G)^0 = I$ the no. of length 0 paths b/w any two vertices u, v

$(A_G)^1$ length 1

$(A_G)^2$ length 2

$(A_G)^k$ length k

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$(A_G)^2 =$

	a	b	c	d
a	0	0+2	1	
b	0	1	1	0
c	0	0	1	1
d	0	1	0	0

Defⁿ: The length K path counting matrix for an n -vertex graph G is the $n \times n$ matrix C s.t

C_{uv} = the no. of length K paths from u to v

Theorem: If C is length K path counting matrix and D is the length m path counting matrix then CD is the length $(K+m)$ path counting matrix.

Corollary: The length K path counting matrix of a digraph G is $(A_G)^K$ for all $K \in \mathbb{N}$

proof: By induction on K

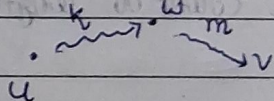
Basis: $K=0$ $A_G^0 = I$ which is trivial

Inductive step: Assume true for $K=n$

$(A_G)^{n+1} = A_G (A_G)^n$ from above then

proof of the theorem:

Any length $= K+m$ path b/w vertices u and v begins with a length K path starting at u and ending at some vertex w followed by a length m path starting at w and ending at v



The no. of such paths $= C_{uw} D_{wv}$

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So no. of all $(k+m)$ length paths from u to v are

$$= \sum_{w=0}^{n-1} C_{uw} D_{wv} = (CD)_{uv}$$

$\therefore CD$ is $(k+m)$ length path containing matrix

To compute the shortest path from u to v :

A_0 Stop when you see the entry $(A_k)_{uv} > 0$

A_k^2 k is the length of shortest path from u to v .

A_3

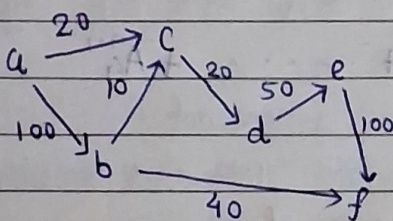
$\vdots$

A_{n-1}

Weighted graphs.

the edges have weights (or ~~some~~ some cost of walking on them)

examples.



$A_0 =$

Its adjacency matrix can be represented as following:

	a	b	c	d	e	f
a	0	100	20	0	0	0
b	0	0	10	0	0	40
c	0	0	0	20	0	0
d	0	0	0	0	50	0
e	0	0	0	0	0	100
f	0	0	0	0	0	0

9th Relations:

Note: ① Transitive closure of a relation R on set A is the smallest transitive relation on A that contains R .

② if transitive closure is also reflexive then it's called reflexive transitive closure.

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Consider the semiring: $(K, +, \cdot, 0, 1)$ ✓ commutative monoid
↑
monoid.

Examples:

① $(\mathbb{N}, +, \cdot, 0, 1)$

$\Rightarrow (A_G)^k_{uv}$ counts the no. of distinct paths of length k from u to v

② $(\mathbb{B}, \vee, \wedge, 0, 1)$

$\Rightarrow (A_G)^k_{uv} = \begin{cases} 1 & \text{if there is a path from } u \text{ to } v \\ & \text{(of length } \leq k) \\ 0 & \text{otherwise} \end{cases}$

③ $(\mathbb{R}^+, \min, +, \infty, 0)$

: generally used for weighted graph

$\Rightarrow (A_G)^k_{uv}$ is minimum weight (or cost) of a path of length k from u to v

What does A_G^* mean?

$$A_G^* = I + A_G + A_G^2 + \dots + A_G^{n-1}$$

① $(\mathbb{N}, +, \cdot, 0, 1)$

$\Rightarrow (A_G^*)_{uv}$ gives ^{total} no. of paths from u to v
paths of any path length from 0 to $(n-1)$

② $(\mathbb{B}, \vee, \wedge, 0, 1)$

$\Rightarrow (A_G^*)_{uv} = 1$ if there exist at least one path from u to v of any length

③ $(\mathbb{R}^+, \min, +, \infty, 0)$

$\Rightarrow (A_G^*)_{uv}$ represents cost of shortest path (minimum cost) from u to v

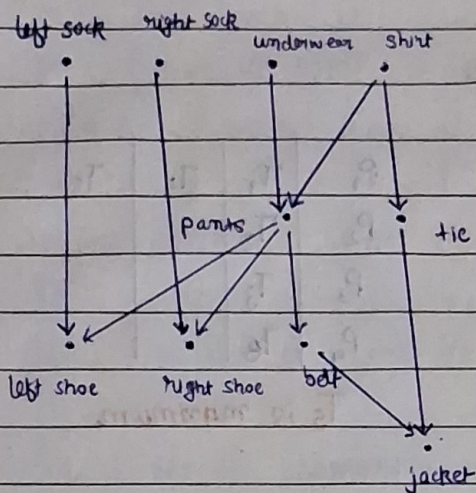
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Direct Acyclic Graphs (DAG)

Directed graphs with no cycle.

Scheduling problem:



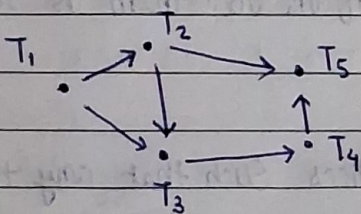
Given a set of tasks, along with a set of constraints.

Specifying that certain tasks depends on others.

Basic problem:

— find an order to execute the task ~ **Topological sorting**

for example:



a possible top

a possible topological sort is:

T_1, T_2, T_3, T_4, T_5

defⁿ: A topological sort of a finite DAG is a list of all the vertices such that each vertex v appears earlier in the list than every other vertex reachable from v .

defⁿ: A vertex v of DAG D is minimum iff every other vertex is reachable from v .

defⁿ: **minimal:** A vertex v is minimal iff v is not reachable from any other vertex.

Theorem: Every finite DAG has a topological sort.

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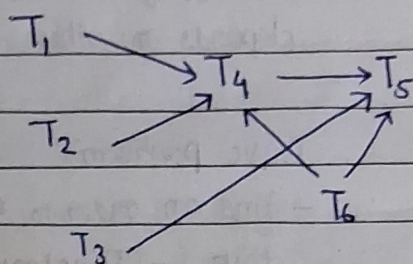
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DAG $\leftrightarrow$ finite partial order

$v_i \leq v_j$ in PO

$\Rightarrow v_i < v_j$ in TO

Parallel Task Scheduling



	T_1	T_4	T_5
P_1			
P_2			
P_3			
P_4			

T_5 is maximum

Defⁿs

- Two vertices in DAG comparable if one of them is reachable from the other.
- A chain in a DAG is a set of vertices such that any two of them are comparable.

maximum element: A vertex in a chain that is reachable from any other vertex in the chain is called the maximum elt of the chain.

The time taken to schedule tasks with an unbounded no. of processors is at least as large as the no. of vertices in any chain.

i.e. # processor $\geq$ max^m length chain's length

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Critical path: largest chain is also known as critical path.

partition of Set A: Set of nonempty subsets of A called the blocks of the partition such that every element of A is in exactly one block.

eg. $\{a, b, c, d, e\}$

$\{a, c\} \{b, e\} \{d\}$ } 3 blocks

parallel schedule for DAG: ~~D is a partition~~

D is a partition of $V(D)$ into blocks $A_0, A_1, \dots$ such when $j < k$ no vertex in A_j is reachable from any vertex in A_k .

① A_k : set of elements scheduled at step k .

② time to schedule = no. of blocks

③ max^m number of elements scheduled at any step.

④ max^m number of elements scheduled at any step = no. of processors required by the schedule.

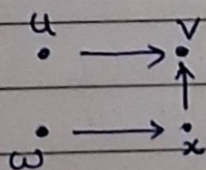
Defⁿ:

- A largest chain ending at vertex v is called critical path to v .
- no. of elements less than v in the chain is called depth of v .

in any parallel schedule there must be at least depth(v) steps before task v can be started.

∃ simple scheduling that completes every task in its minimum no. of steps.

~ **Greedy scheduling**



depth(v) = 2

Theorem: A minimum time schedule for finite DAG consists of sets $A_0, A_1, \dots$ where

$$A_k = \{a \in V(D) \mid \text{depth}(a) = k\}$$

Corollary: Parallel time = Size of Critical path

Defⁿ: An antichain in a DAG is a set of vertices such that no two elements in the set are comparable.

$\{T_1, T_2, T_3, T_6\}$ — antichain

$\{T_2, T_6\}$

$\{T_1, T_3, T_5\}$ not an antichain

Corollary: In a DAG D if the size of the largest chain is t then $V(D)$ can be partitioned into t antichains.

proof:

Let the ~~ex~~ antichains be the sets

$$A_k = \{v \in V(D) \mid \text{depth}(v) = k\}$$

It is easy to show that each A_k is an antichain.

Dilworth Lemma

$\forall t > 0$ every DAG with n vertices must have either a chain of size $> t$

(OR) an antichain of size at least $\frac{n}{t}$

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Proof: Assume no chain of size $> t$ exists.
Let l be the size of the largest antichain.

if we make parallel scheduling (like prev proof)

antichains = size of largest chain $\leq t$

no. of vertices = $n \leq l \cdot (\# \text{ antichains}) \leq l \cdot t$

$$\therefore n \leq l \cdot t$$

$$l \geq \frac{n}{t}$$

$\therefore \exists$ an antichain of ~~length~~ length $\geq \frac{n}{t}$